Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science & Engineering

MCA (2nd year): MST | Course Code: MCA302
Oct 03, 2023 | Course Name: Object Oriented Programming
Time: 2 Hours, M. Marks: 35 | Name of Faculty: Dr. Sujata

Note: Attempt all questions Assume missing data, if any, suitably

Q1: Write a program in C++ to

I. Create a base class **shape**. Use this class to store two double type values(val1 and val2) that could be used to compute the area of figures.
II. Derive two specific classes called **triangle** and **rectangle** from the base class **shape**. Define a member function *get_data()* of the base class to interactively initialize base class data members and another member function *display_area()* to compute and display the area of figures. Make *display_area()* as a pure virtual function and redefine this function in the derived classes to suit their requirements.
III. Using these three classes, design a program that will accept the dimensions of a triangle and a rectangle interactively from the user, and display the area by using an array of base class pointers(or use multiple pointers to the base class objects) to the derived classes.
IV. Define each member function outside the class.

The two values given in the input will be treated as lengths of two sides in case of rectangle and as base and height in the case of the triangle and used as follows:
Area of rectangle = val1 * val2, Area of triangle = 0.5* val1 *val2 [7]

Q2 (a): What is the need of the access specifier "protected" in C++? [2]
(b): What is the scope resolution operator? Explain its three different usages with examples for each. [4]
(c): Explain the difference between early and late binding in C++? [2]

Q3: A theater sells seats for shows and needs a system to keep track of the seats they have sold tickets for. Define a class for a type called ShowTicket. The class should contain private member variables for the row, seat number and whether a ticket has been sold or not. Your class should include the following member functions:

- a default constructor that initializes the row and seat number to 0 and the sold status to false
- an overloaded constructor which accepts as arguments the row and seat number and sets the sold status to false
- a member function to check if the ticket has been sold
- a member function to update the ticket status to sold
- a member function to print the row, seat number and sold status
- a destructor.

Embed your class definition in a test program which creates some ShowTicket objects, sets some tickets as sold, and prints each of them out. [7]

Q4: Create a class Armstrong and include startnum and endnum as data member and also include member function findArmstrongNumbers() to find all the Armstrong numbers between startnum and endnum. If the sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number. [6]

Q5 (a): There cannot be two variables/two functions with the same name in the same scope. Using namespaces, we can create two variable or member functions having the same name. Justify this statement with the help of an example considering variables and member functions both. [4]
(b): Is destructor overloading possible? If yes then explain and if no then why? [3]